

Midterm

Phil 305 — Propositional and Modal Logic

Part 1: S5

Use the simplified rules for **S5** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S5**.

1. $\Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$
2. $\Diamond\Box p \rightarrow \Box\Diamond p$
3. $\Diamond(\neg p \vee r) \rightarrow (\Box(p \vee q) \rightarrow \Diamond(q \vee r))$
4. $(\neg p \vee \neg\Box\Box p) \rightarrow \neg\Box p$

Part 2: K

Use the rules for **K** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **K**.

5. $(\Diamond p \wedge \Diamond q) \rightarrow \Box(p \vee q)$
6. $\Diamond(p \rightarrow p) \vee \Diamond\neg(q \rightarrow q)$
7. $\Box(p \rightarrow \neg q) \rightarrow \Box(q \rightarrow \neg p)$

Part 3: S4

Use the rules for **S4** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S4**. Remember **S4** is the logic **KT4**, so it has the rules for **T** and **4** as well as **K**.

8. $\Box(p \rightarrow q) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond q)$
9. $(p \wedge \Diamond\Box p) \rightarrow \Box p$
10. $\Diamond\Box\Diamond\Box p \rightarrow \Diamond\Box p$